

Numerical computation of the time-averaged (Schindler-Bukov) energy of Floquet eigenstates for the kicked top

1 Model and Floquet setup

We consider a spin- J system (Hilbert space dimension $d = 2J + 1$) with Hamiltonian

$$\hat{H}(J, \alpha, k, t) = \alpha \hat{J}_z + \frac{k}{2J} \hat{J}_x^2 \sum_{n=0}^{\infty} \delta(t - n\tau), \quad (1)$$

where τ is the driving period, α and k are real parameters, and $\hat{J}_{x,y,z}$ are angular-momentum operators.

1.1 One-period evolution operator

Choose the stroboscopic origin $t = 0^-$ *just before* a kick. Then, for $0 < t < \tau$ the Hamiltonian is simply $\alpha \hat{J}_z$, and at $t = \tau$ there is an instantaneous kick generated by $\frac{k}{2J} \hat{J}_x^2$.

Define

$$\hat{U}_0 := e^{-i\alpha\tau\hat{J}_z}, \quad (2)$$

$$\hat{U}_K := e^{-i\frac{k}{2J}\hat{J}_x^2}. \quad (3)$$

With the above convention (free evolution then kick at the end of the period),

$$\hat{F} \equiv \hat{U}(\tau^+, 0^-) = \hat{U}_0 \hat{U}_K. \quad (4)$$

Different stroboscopic conventions correspond to cyclic permutations (or a symmetric splitting), and change the micromotion while leaving quasienergies invariant modulo $2\pi/\tau$. The *time-averaged energy* defined below is convention-dependent; fix a convention and use it consistently.

1.2 Floquet eigenproblem

Compute eigenpairs of $\hat{F}$:

$$\hat{F} |\phi_n\rangle = e^{-i\phi_n} |\phi_n\rangle, \quad n = 1, \dots, d, \quad (5)$$

where $\phi_n \in (-\pi, \pi]$ can be chosen as the principal eigenphases and the quasienergies are $\varepsilon_n = \phi_n/\tau$ modulo $2\pi/\tau$. Note that $|\phi_n\rangle \equiv |\phi_n[t_0 = 0^-]\rangle$.

2 Time-averaged energy of a Floquet eigenstate

2.1 Definition

Given a Floquet eigenstate $|\phi_n\rangle$ (at $t = 0^-$), define its *time-periodic Floquet state* within one period,

$$\hat{U}(t_0 + \tau, t_0) |\phi_n[t_0]\rangle = e^{-i\phi_n} |\phi_n[t_0]\rangle. \quad (6)$$

This formula agrees with our definition above for $\hat{F} \equiv \hat{U}(\tau^+, 0^-)$ that you have been using so far. We can also change the gauge of

The time-averaged energy is the cycle average of the instantaneous energy expectation:

$$\bar{E}_n = \frac{1}{\tau} \int_0^\tau dt \langle \phi_n[t] | \hat{H}(t) | \phi_n[t] \rangle = \langle \phi_n[0] | \left[\frac{1}{\tau} \int_0^\tau dt \hat{U}^\dagger(t, 0) \hat{H}(t) \hat{U}(t, 0) \right] | \phi_n[0] \rangle. \quad (7)$$

Define $\bar{V} = \frac{1}{\tau} \int_0^\tau dt \hat{U}^\dagger(t, 0) \hat{H}(t) \hat{U}(t, 0)$, we can calculate this explicitly since we have a kick drive where we know the Floquet unitary exactly.

2.2 Calculation of $\bar{V}$

This expression can be simplified significantly. One can show that $\bar{V}$ is given by

$$\bar{V} = \frac{k}{2J\tau} \hat{J}_x^2 + \alpha \hat{U}_K^\dagger \hat{J}_z \hat{U}_K,$$

where $\hat{U}_K$ was defined earlier. This follows from the fact that for any time greater than 0 but less than τ , the Hamiltonian is time-independent.

Note this is almost the same as just averaging the Hamiltonian, but not quite since we have the additional $\hat{U}_K$ sandwiching the $\hat{J}_z$ operator.

Hence, the average energies are the expectation values

$$\bar{E}_n = \langle \phi_n | \frac{k}{2J\tau} \hat{J}_x^2 + \alpha \hat{U}_K^\dagger \hat{J}_z \hat{U}_K | \phi_n \rangle.$$

This is subtly different than the g_n that we had already written down since we also have the kick operators sandwiching it. Not clear that they will give the same quantity.

What is also interesting to note is that we can shift our convention to $0^+ \rightarrow \tau^+$, such that the kick comes at the end,

$$\hat{F}' = U_K U_0,$$

with different eigenstates $|\phi'_n\rangle$. This is very important since we cannot mix notation with prime and unprime.

In this convention, we find instead that

$$\bar{E}_n = \langle \phi'_n | \alpha \hat{J}_z + \frac{k}{2J\tau} \hat{U}_K^\dagger \hat{J}_x^2 \hat{U}_K | \phi'_n \rangle.$$

This produces equivalent values, but we need to make sure to use the new Floquet eigenstates $|\phi'_n\rangle$.

3 Checks

As we talked about earlier, one can check this by calculating $|\phi_n[t]\rangle$ as the eigenvectors of $U(T+t, t)$. This unitary can be found as a rotation using

$$\hat{U}(T+t, t) = \exp(i\alpha \hat{J}_z(\tau - t)) \hat{F} \exp(-i\alpha \hat{J}_z(\tau - t)). \quad (8)$$